

**Mohammed Reabai J 250701435-2025-CSE** ▾**M2****Started on** Monday, 24 August 2026, 6:50 PM**State** Finished**Completed on** Monday, 24 August 2026, 6:59 PM**Time taken** 9 mins 19 secs**Marks** 1.00/1.00**Grade** **10.00** out of 10.00 (**100%**)

Question 1 | Correct | Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void function (int n)
```

```
{
```

```
    int i= 1;
```

```
    int s =1;
```

```
    while(s <= n)
```

```
    {
```

```
        i++;
```

```
        s += i;
```

```
    }
```

```
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

For example:

Input	Result
9	12

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2  int main(){
3      int n;
4      int i=1;
5      int s=1;
6      int count=0;
7      scanf("%d",&n);
8      while(s<=n){
9          i++;
10         count++;
11         s+=i;
12         count++;
13         count++;
14     }
15     count+=3;
16     printf("%d",count);
17     return 0;
18 }
```

	Input	Expected	Got	
✓	9	12	12	✓
✓	4	9	9	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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